

Topic:Basic Oracle Net ArchitectureOracle Net:

⇒ Oracle net, also known as Oracle net services, is a suit of software protocols and components designed by Oracle Corporation to facilitate network communication between clients and servers in Oracle.

⇒ Serve as the middle layer responsible for establishing, managing and securing connections b/w DB clients and servers.

Purpose of Oracle net:

- o Connects to distributed database
- o Facilitates data sharing b/w different databases.
- o Supports different servers, OS and communication protocols.

Definition of Oracle net architecture:

Also known as Oracle network architecture.

⇒ it is the framework and components responsible for facilitating the communication b/w clients and servers in Oracle database environments.

⇒ it encompasses various software components, protocols and the configuration settings designed to establish, manage and secure network connections.

⇒ Break down into:

- o Client-server Communication
- o Client Components
- o Server Components
- o Oracle Net services
- o Listener
- o Configuration files
- o Protocols
- o Security

Overview of Oracle Net

Distributing database workload with Oracle Net:

- ⇒ Oracle Net helps distribute the workload of database application.
- ⇒ Instead of one server handling both CPU and I/O tasks, the workload is shared.

★ Client Server Configuration:

How it works?

Client:

- Runs the application that initiates database requests.

Server:

- manages the database and processes SQL queries.

Benefits:

- Reduces the load on server.
- Client handles data presentation.
- Server focuses on database queries.

Fat-client architecture

Description:

- Client performs significant processing tasks.
- Higher resource requirements for clients.

Cost impact:

- Though workstation costs have dropped, the overall cost to a company can still be high.

Thin-client Configuration (Three-tier architecture)

How it works?

Client:

- Connects to the application server.

Application Server:

- Runs the application code using java applets.

Database Server:

- Executes SQL queries.

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Benefits:

- o Low resource requirement for clients.
- o Centralized maintenance on the application server.
- o High resources but lower maintenance for the database server.

Communication flow in thin-client architecture:

Client actions:

- o Connects to the application server.
- o Downloads display management code.
- o Sends the database requests via the application server.

Server actions:

- o Application server forwards requests to the database server.
- o Database server executes SQL statements and returns

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the result via the application server.

Servers/Server Configurations

- o Separate servers share data with each other.
- o Servers can be physically isolated but logically connected.

Typical usecases:

- o Corporate head/quarter servers communicating with departmental servers.

Operation:

- o A server can act as client when sending database requests to another server.

Oracle net Capabilities

Database Communication:

- o Allows database requests to be passed between different servers.

Network Transparency:

- Users do not need to know the physical location of data.

Object identification in Databases:Standard database object:

- Identified by its owner and name.

Distributed database object:

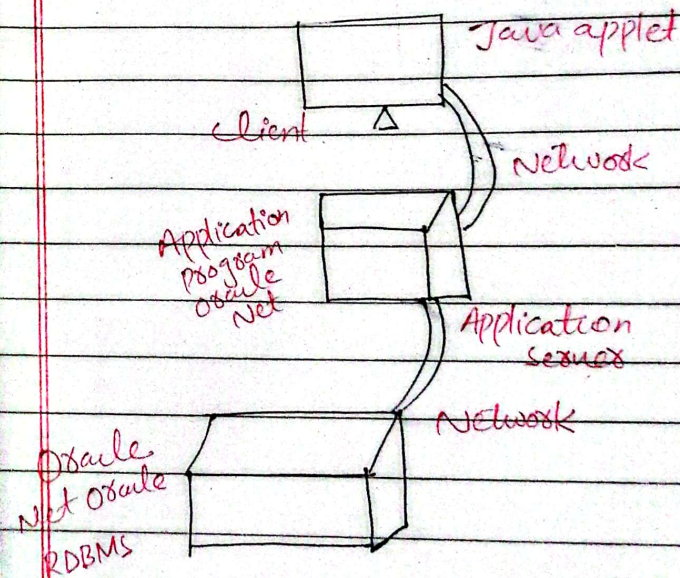
- Additional identifiers: instance name and server name.
- Forms a global object name

Transparent Network Substrate (TNS)Purpose:

- Resolves server connectivity issues.

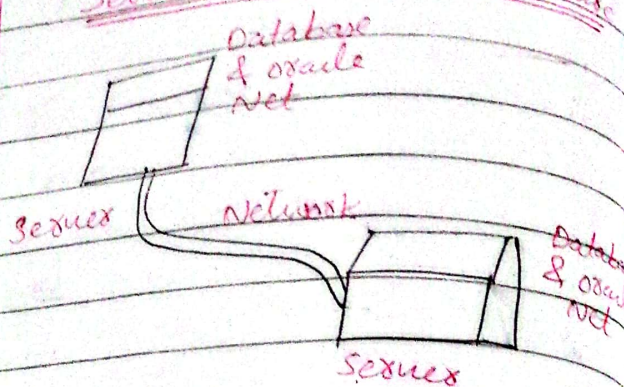
Function:

- uses configuration files on both client and server sides.
- manages connection even if different communication protocols are used.

Diagrams:Thin-client Architecture:

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Server-Server Architecture



collection of computers, servers, mainframes or other devices, interconnected to one another allowing the sharing of data.

Types of Networks

⇒ There are various types of network that can be used to facilitate communication between clients and servers.

⇒ These networks can vary based on geographical scope, security requirements and connectivity methods.

main types

LAN: for short-distance, high speed communication within buildings or campuses.

WAN: for long-distance connectivity across cities or countries.

VPN: Secure communication over public network.

Internet: Global access with necessary security measures.

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Extranet:

- Controlled access for external users.

Intranet:

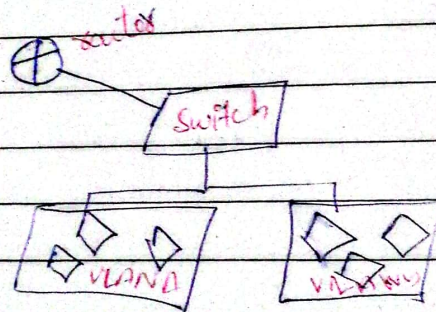
- Private, internal communication within an organization

SAN: (Storage area network)

- High speed network for connecting to large scale storage systems.

MAN:

VLAN: Allow multiple LANs & isolated devices to communicate via wireless internet



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Oracle Net Services

- establishing connection
- data transfers
- location transparency
- security & scalability

Definitions

⇒ Critical component of Oracle DB environment, providing the infrastructure necessary for network communication between clients and servers.

⇒ It includes a suit of software components and tools that facilitates the establishment, management and security of these network connections.

⇒ **Go** After a network session is established, Oracle Net act as data couriers for both the client and database servers.

⇒ Oracle Net is connected or located on each computer that need to talk to the database servers.

⇒ On client computers:

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Oracle net is background component for application connection to the database.

⇒ On the database server:

Oracle net includes an active process called Oracle net listener, which is responsible for coordinating connections between database and external application.

Components

Oracle net:

⇒ It is networking stack, that runs on both client devices and Oracle databases in order to maintain connection and messaging between client applications and server databases.

it is formally called SQL*Net

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The Listener:

⇒ This process on server detects incoming request from client for connection and manages network traffic once client has connected to server database.

⇒ It is server-side process that listens for incoming client connection requests and routes them to appropriate database server.

Listener Configuration:

⇒ Defined in the "listener.ora" file specifying the network protocols and addresses the listener will use, and the service it will manage.

Oracle Connection Manager:

⇒ CMAN acts as a lightweight router for Oracle net packets.

⇒ Handles the lifecycle of network connections b/w clients and servers, ensuring efficient & reliable connectivity.

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⇒ Key features include:

- Connection establishment
- Connection pooling - reusing existing connections
- Connection Termination

Oracle Net manager:

⇒ Oracle net manager, a GUI tool, configures Oracle net services for an Oracle home on a local client or server host.

Protocols supported:

⇒ Oracle net services supports various network protocols to ensure compatibility with different network environments.

- TCP/IP
- UDP/IP
- IPC

Configuration files:

Oracle net services relies on several configuration files to manage network

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settings and parameters.

tnsnames.ora:

⇒ maps services names to network addresses.

⇒ Allowing clients to connect to database services using a simple file.

sqlnet.ora:

⇒ Contains global settings for Oracle net services.

listener.ora:

⇒ Configures the Oracle net listener, specifying its listening addresses, protocols, and the services it manages.

Diagnostic and monitoring Tools:

⇒ Provides various tools:

- tnsping
- Oracle net manager
- Oracle enterprise manager
- Oracle net Configuration assistant
 - ↳ enable listeners & user to configure naming methods
- Oracle net trace assistant
- Command line control

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Why optimize oracle net?

System performance
High availability
Network scalability
Network manageability
Network security

Oracle Net 11g features:

⇒ Performance & Scalability
⇒ High availability & manageability
Network security

Note: for understanding.

Features of Oracle Net Services:

1. Connectivity
↳ Three types
2. Manageability
3. Performance & Scalability
↳ Through connection pooling & session multiplexing
4. Network security
↳ Firewall AC
↳ Protocol AC

Oracle Shared Server

Definition:

- o Also known as multi-threaded server.
- o Allows multiple user processes to share a small number of server processes instead of having their own dedicated server process, improving resource utilization and scalability.

Key Concepts:

Dispatches processes: → act as ^{listener}

- o Handles incoming client requests and route them to available shared server processes.

The client - establish connection with dispatch using TCP/IP protocol

Shared server processes: ^{Pool} lightweight

- o Execute database requests on behalf of user processes.

- each shared server process can switch b/w different connections.

Request and response queues:

- o Buffers where requests are placed and picked up by server process that is free from which responses are retrieved.

Shared server pool: A pool of the DB server process on

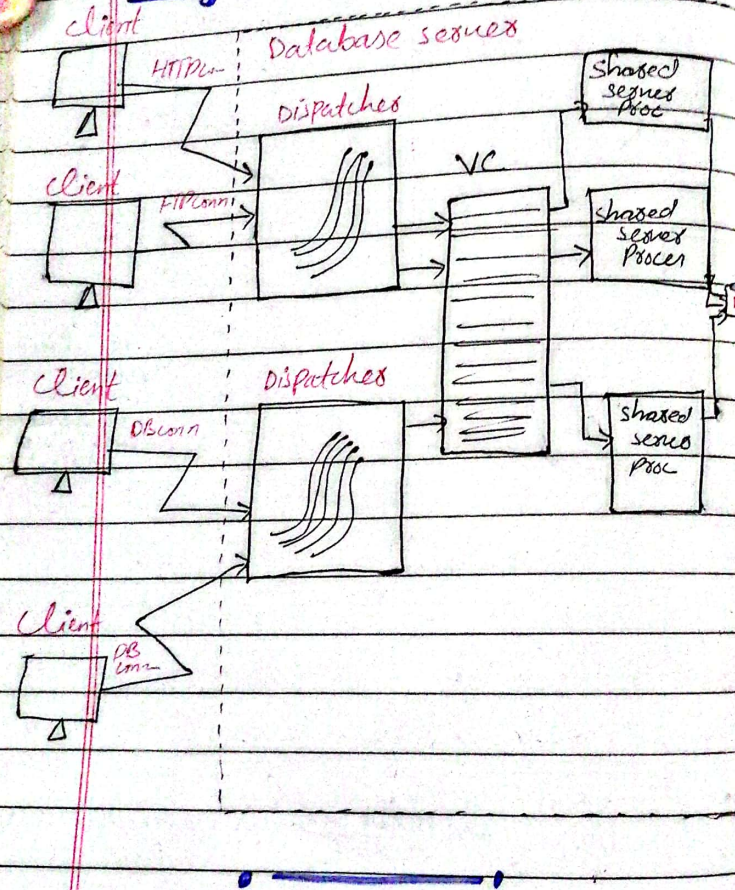
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Benefits

- Reduced server consumption
- Scalability
- Connection multiplexing

Diagram:



Q. What is Connect descriptors?

Definition: Each net services name is associated with a connect descriptor is a connect descriptor

Configuration that tells Oracle net how to connect to a specific database instance on a server. It provides the location of db and the name of the db service.

Components: it specifies the communication protocol, service name, and instance service name, along with hardware connectivity details. Also comprised of one or more protocol addresses of the listener and connect information for the destination.

(DESCRIPTION =

(ADDRESS =

(PROTOCOL = TCP)

(HOST = HQ)

(PORT = 1521))

(CONNECT DATA =

(SERVICE_NAME = LOC))

Define the service name of the DB instance.

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Q. What are Net services Names?

→ Also known as aliases.

Purpose:

Simplifies database connections by using easy-to-remember aliases instead of full names to connect to descriptors.

Storage:

- Stored in the 'tnsnames.ora' file and corresponding connection descriptors.
- Should be present on all servers, clients and App servers.
- The file is located in specific directories based on the OS.

Usage:

- Users can connect to database using these aliases, making database access straightforward & user-friendly.

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Replacing 'tnsnames.ora' with Oracle Internet Directory (OID):

Directory definition:

- electronic database that stores information about various objects.

OID:

- Instead of managing 'tnsnames.ora' on different machines.
- A tool provided by Oracle for centralized storage and management of users, servers, and database information.

Benefits:

- Centralized management of connection details.
- Easier access to update information.
- Simplified administration.